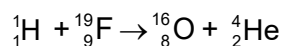


- 7 In a nuclear reaction, a stationary fluorine-19 is bombarded with a proton having a kinetic energy of 5.00 MeV. The following reaction may occur.



The following data may be used for the calculation.

Rest mass of ${}^{19}_9\text{F}$	18.998403 u
Rest mass of ${}_0^1\text{n}$	1.008665 u
Rest mass of ${}_1^1\text{H}$	1.007825 u
Rest mass of ${}_2^4\text{He}$	4.003860 u

- (a) Explain what is meant by

- (i) binding energy of a nucleus,

It is defined as the energy **needed** just to **separate completely** to infinite the nucleons in the nucleus.

Or

It is also defined as energy **released** when a **nucleus is formed** from infinite by putting all its constituent nucleons together

- (ii) mass defect of a nucleus.

Mass defect is defined as the difference between the total mass of the nucleons of the atom taken separately and the mass of the nucleus.

.....[1]

- (b) Calculate the binding energy per nucleon, in MeV, for fluorine-19.

$$\text{Mass defect} = (\sum m_p + \sum m_n) - m_{\text{nucleus}}$$

$$= (9(1.007825\text{u}) + (19-9)(1.008665\text{u}) - 18.998403\text{u})$$

$$= 0.158672 \text{ u}$$

$$= 2.633955 \times 10^{-28} \text{ kg} \quad [1]$$

$$\text{Binding energy per nucleon} = \frac{\text{mass defect} \times c^2}{19}$$

$$= \frac{(2.633955 \times 10^{-28}) \times (3.0 \times 10^8)^2}{19}$$

$$= 1.247663 \times 10^{-12} \text{ J} \quad [1]$$

$$= 7.80 \text{ MeV} \quad [1]$$

binding energy per nucleon = MeV [3]

- (c) Use the answer in (b) and the following data to determine the total kinetic energy of the products. Assume no photon is emitted in this reaction.

	Binding energy per nucleon/MeV
Oxygen-16	7.72
Helium-4	6.82

$$\begin{aligned}
 \text{Total energy} &= (\text{total BE})_{\text{products}} - (\text{total BE})_{\text{reactants}} + \text{input energy} \\
 &= [(7.72)(16) + (6.82)(4)] - (7.80)(19) + 5.00 \\
 &= 7.60 \text{ MeV} \\
 &= 1.22 \times 10^{-12} \text{ J}
 \end{aligned}$$

total kinetic energy = J [2]

- (d) Determine the rest mass of $^{16}_8\text{O}$.

By conservation of energy,

$$(\Sigma M)_{\text{reactants}} \times c^2 + KE_{\text{reactants}} = (\Sigma M)_{\text{products}} \times c^2 + KE_{\text{products}}$$

$$\Rightarrow [(18.998403 \text{ u}) + (1.007825 \text{ u})] c^2 + (5.00 \times 10^6)(1.60 \times 10^{-19})$$

$$= (4.003860 \text{ u})(c^2) + M_{\text{O}}c^2 + 1.22 \times 10^{-12} \text{ J}$$

$$\text{Rest mass} = 2.655926 \times 10^{-26} \text{ kg} = 2.66 \times 10^{-26} \text{ kg}$$

Alternatively,

Binding energy of Oxygen = mass defect $\cdot c^2$.

$$(7.72)(16) \times 1.60 \times 10^{-19} \times 10^6 = (\text{mass of 8 protons} + 8 \text{ neutrons} - \text{rest mass of oxygen})(3.0 \times 10^8)^2$$

$$\text{rest mass of oxygen} = 8 \times (1.007825 + 1.008665)(1.66 \times 10^{-27}) - 2.195911 \times 10^{-28}$$

$$= 2.655940 \times 10^{-26} \text{ kg} = 2.66 \times 10^{-26} \text{ kg}$$

[Total : 9]

